Univariate Exploratory Data Analysis (EDA)

In [9]:

```
# Upload dataset and read it into a pandas DataFrame
%pip install pandas
import pandas as pd

path = r"C:\Users\ramir\OneDrive\TMU Capstone\HR-Employee-Attrition-Dataset.csv"
df = pd.read_csv(path)
df.head()
```

```
Requirement already satisfied: pandas in .\.venv\Lib\site-packages (3.0.0)
Requirement already satisfied: numpy>=1.26.0 in .\.venv\Lib\site-packages (from pandas) (2.4.2)
Requirement already satisfied: python-dateutil>=2.8.2 in .\.venv\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in .\.venv\Lib\site-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in .\.venv\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

Out[9]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeC
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	

5 rows × 35 columns

Dataset Overview & Data Integrity Checks

The IBM HR Analytics dataset contains 1470 rows and 35 columns, consisting primarily of int64 numerical variables and object categorical variables. A preliminary integrity review shows:

- No missing values
- No duplicate records
- Consistent datatypes appropriate for HR analytics

This indicates the dataset is clean and ready for exploratory analysis without additional preprocessing.

In [10]:

```
# Number of rows and columns
rows, cols = df.shape
print("Rows:", rows)
print("Columns:", cols)
```

```
Rows: 1470
Columns: 35
```

In [11]:

```
# Check data types of each column
df.dtypes
```

Out[11]:

```
Age                       int64
Attrition                 str
BusinessTravel            str
DailyRate                 int64
Department                str
DistanceFromHome          int64
Education                 int64
EducationField            str
EmployeeCount             int64
EmployeeNumber            int64
EnvironmentSatisfaction   int64
Gender                    str
HourlyRate                int64
JobInvolvement            int64
JobLevel                  int64
JobRole                   str
JobSatisfaction           int64
MaritalStatus             str
MonthlyIncome             int64
MonthlyRate               int64
NumCompaniesWorked        int64
Over18                    str
OverTime                  str
PercentSalaryHike         int64
PerformanceRating         int64
RelationshipSatisfaction  int64
StandardHours             int64
StockOptionLevel          int64
TotalWorkingYears         int64
TrainingTimesLastYear     int64
WorkLifeBalance           int64
YearsAtCompany            int64
YearsInCurrentRole        int64
YearsSinceLastPromotion   int64
YearsWithCurrManager      int64
dtype: object
```

In [12]:

```
# Check for missing/null values
missing_count = df.isnull().sum()
missing_percent = (missing_count / len(df)) * 100
missing_summary = pd.DataFrame({
    'missing_count': missing_count,
    'missing_percent': missing_percent
})

total_missing = missing_count.sum()
print("Total missing values:", total_missing)

if total_missing == 0:
    print("No missing/null values found in the dataset.")
else:
    print("\nMissing values by column:")
    display(missing_summary[missing_summary['missing_count'] > 0].sort_values('missing_count', ascending=False))
    print("\nSample rows with any missing values:")
    display(df[df.isnull().any(axis=1)].head())
```

```
Total missing values: 0
No missing/null values found in the dataset.
```

In [13]:

```
# Check for duplicate records
duplicate_count = df.duplicated().sum()
print("Number of duplicate records:", duplicate_count)

if duplicate_count == 0:
    print("No duplicate records found.")
else:
    print("\nSample duplicate rows:")
    display(df[df.duplicated()].head())
```

```
Number of duplicate records: 0
No duplicate records found.
```

Univariate Distribution of Numerical Features

Age

Displays a roughly normal distribution, suggesting a balanced mix of early-career and mid-career employees.

Right-Skewed Features (Income & Tenure Variables)

Several variables—including MonthlyIncome, TotalWorkingYears, YearsAtCompany, and YearsInCurrentRole—show right-skewed distributions:

- Most employees fall at the lower end of these ranges
- A smaller group extends into much higher values
- This pattern often reflects organizational hierarchy (many junior staff, fewer senior leaders)

Right-skewness is important because it may influence model assumptions and amplify the effect of high-value outliers.

Low-Variance or Non-Informative Features

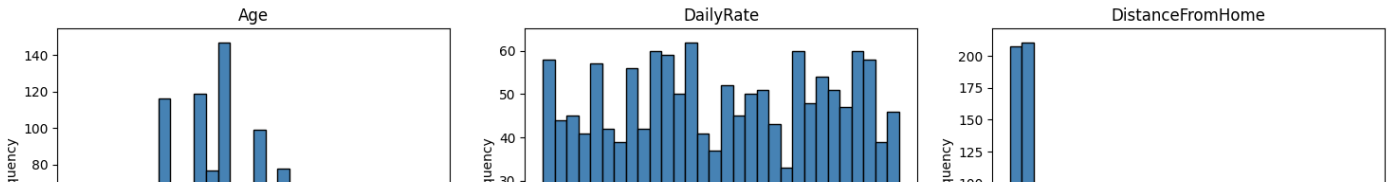
The following features show zero or near-zero variance, making them candidates for removal during dimensionality reduction:

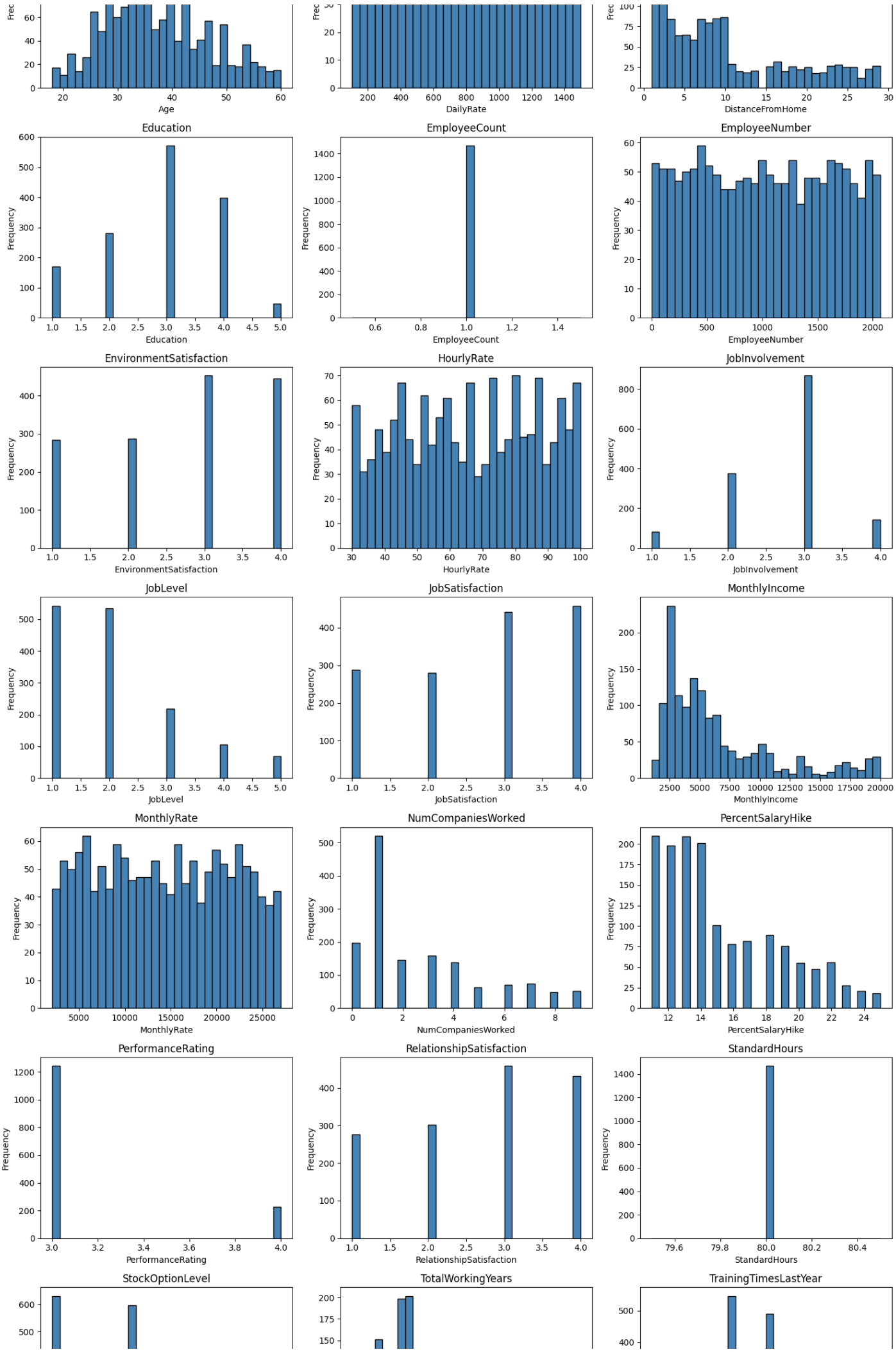
Feature	Issue	Interpretation
EmployeeCount	Constant value	No analytical value
StandardHours	Constant value	No variation in working hours
Over18	Constant "Y"	Regulatory field, not predictive
EmployeeNumber	Unique identifier	Should be removed before modeling
PerformanceRating	Only values 3 and 4	Unusually compressed distribution; lacks predictive value

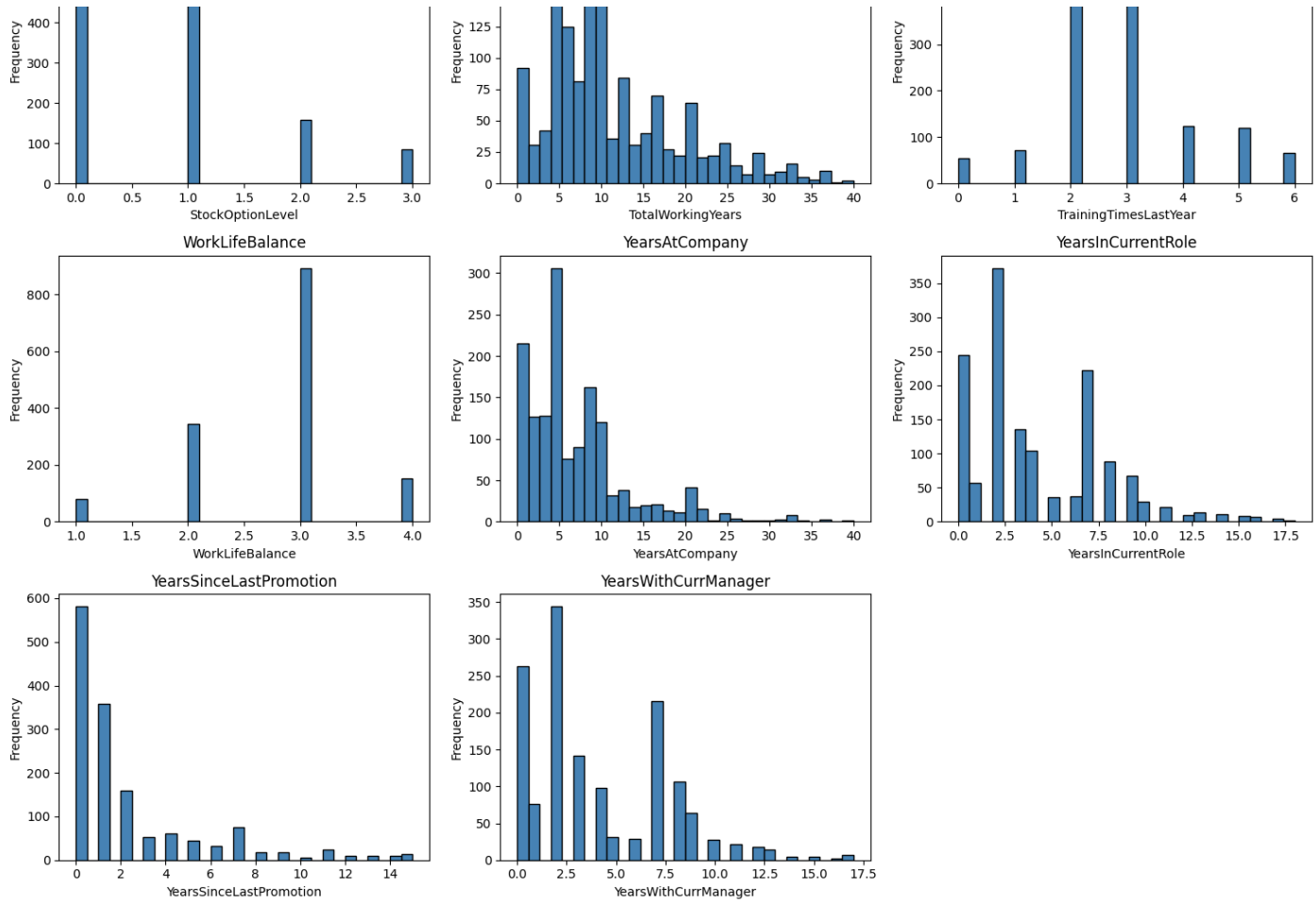
These features do not contribute meaningful signal and can be safely dropped.

In [16]:

```
# Histograms for all numeric features (4 plots per row)
import matplotlib.pyplot as plt
num_cols = df.select_dtypes(include="number").columns

plots_per_row = 3
n_plots = len(num_cols)
n_rows = (n_plots + plots_per_row - 1) // plots_per_row

fig, axes = plt.subplots(n_rows, plots_per_row, figsize=(5 * plots_per_row, 3.8 * n_rows))
axes = axes.flatten()

for i, col in enumerate(num_cols):
    axes[i].hist(df[col].dropna(), bins=30, color="steelblue", edgecolor="black")
    axes[i].set_title(col)
    axes[i].set_xlabel(col)
    axes[i].set_ylabel("Frequency")

# Hide any unused subplots
for j in range(i + 1, len(axes)):
    axes[j].axis("off")

plt.tight_layout()
plt.show()
```




Age
DailyRate
DistanceFromHome
Education
EmployeeCount
EmployeeNumber
EnvironmentSatisfaction
HourlyRate
JobInvolvement
JobLevel
JobSatisfaction
MonthlyIncome
MonthlyRate
NumCompaniesWorked
PercentSalaryHike
PerformanceRating
RelationshipSatisfaction
StandardHours
StockOptionLevel
TotalWorkingYears
TrainingTimesLastYear
Frequency



Outlier Detection via Boxplots

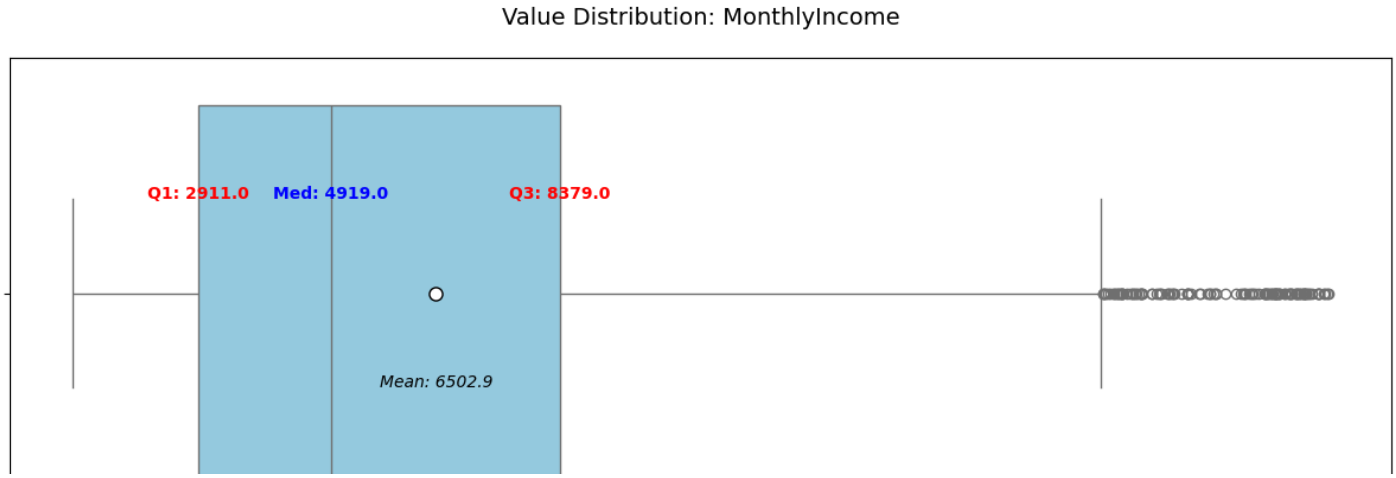
Boxplots reveal several high-value outliers, each with meaningful HR interpretations:

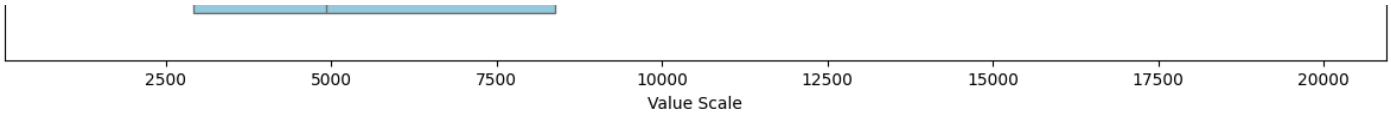
Attribute	Outlier Type	Interpretation
MonthlyIncome	High	Outliers represent senior managers/directors earning $16,500+
TotalWorkingYears	High	28+ year veterans with typically low attrition risk
YearsAtCompany	High	20+ year "lifers" who stayed through multiple organizational cycles
YearsInCurrentRole	High	12–14+ years in the same role suggests specialization or stalled mobility
YearsWithCurrManager	High	10–12+ years with the same manager indicates strong stability
NumCompaniesWorked	High	8–9 prior employers suggests "job hopping," often linked to higher attrition
TrainingTimesLastYear	High/Low	Outliers include employees with 0 (possible neglect) or 6 (specialized/remedial) trainings

These outliers are not errors—they represent meaningful employee subgroups.

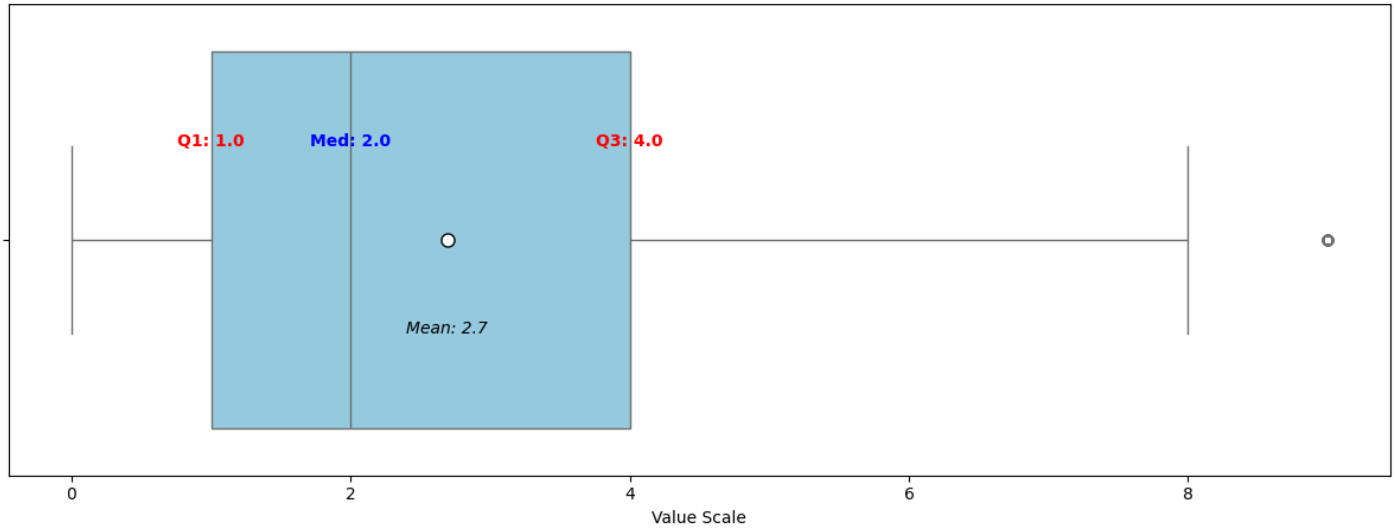
In [ ]:

```python
# Boxplots of numeric features - outlier analysis
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Calculate stats and identify outliers
stats = df.describe().T
stats['Q1'] = stats['25%']
stats['Q2_Median'] = stats['50%']
stats['Q3'] = stats['75%']
stats['IQR'] = stats['Q3'] - stats['Q1']

# 2. Logic for Outlier Detection
stats['outlier_check'] = (stats['max'] > (stats['Q3'] + 1.5 * stats['IQR'])) | \
                         (stats['min'] < (stats['Q1'] - 1.5 * stats['IQR']))

outlier_cols = stats[stats['outlier_check'] == True].index.tolist()

# 3. Create subplots
n_cols = len(outlier_cols)
if n_cols > 0:
    fig, axes = plt.subplots(n_cols, 1, figsize=(12, 5 * n_cols))
    if n_cols == 1: axes = [axes]

    for i, col in enumerate(outlier_cols):
        # showmeans=True adds a marker for the average
        sns.boxplot(x=df[col], ax=axes[i], color='skyblue', showmeans=True,
                    meanprops={"marker":"o", "markerfacecolor":"white", "markeredgecolor":"black", "markersize":"8"})

        # Get the specific values for labeling
        q1, q2, q3 = stats.loc[col, ['Q1', 'Q2_Median', 'Q3']]
        mean_val = stats.loc[col, 'mean']

        # 4. Add Numeric Labels above the box
        # format(val, '.1f') rounds to 1 decimal place
        label_y_offset = -0.2  # Adjusts height of text above the box

        axes[i].text(q1, label_y_offset, f'Q1: {q1:.1f}', ha='center', fontweight='bold', color='red')
        axes[i].text(q2, label_y_offset, f'Med: {q2:.1f}', ha='center', fontweight='bold', color='blue')
        axes[i].text(q3, label_y_offset, f'Q3: {q3:.1f}', ha='center', fontweight='bold', color='red')
        axes[i].text(mean_val, 0.2, f'Mean: {mean_val:.1f}', ha='center', fontstyle='italic', color='black')

        axes[i].set_title(f'Value Distribution: {col}', fontsize=14, pad=20)
        axes[i].set_xlabel('Value Scale')

    plt.tight_layout()
    plt.show()

# 5. Summary Table
print(stats.loc[outlier_cols, ['mean', 'Q1', 'Q2_Median', 'Q3', 'max']])
```

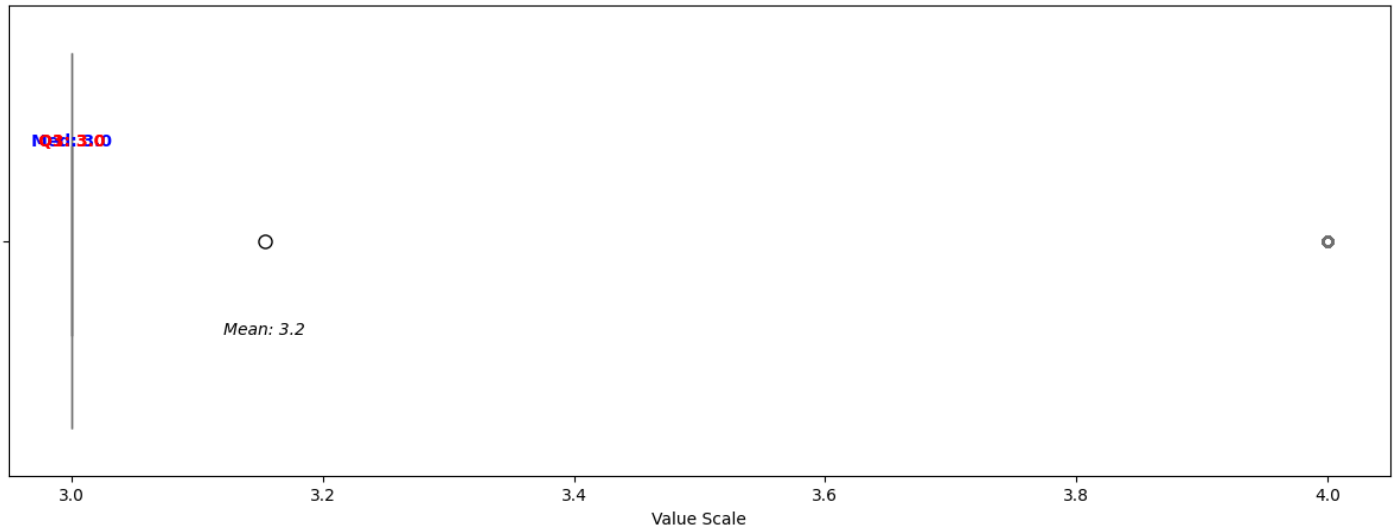




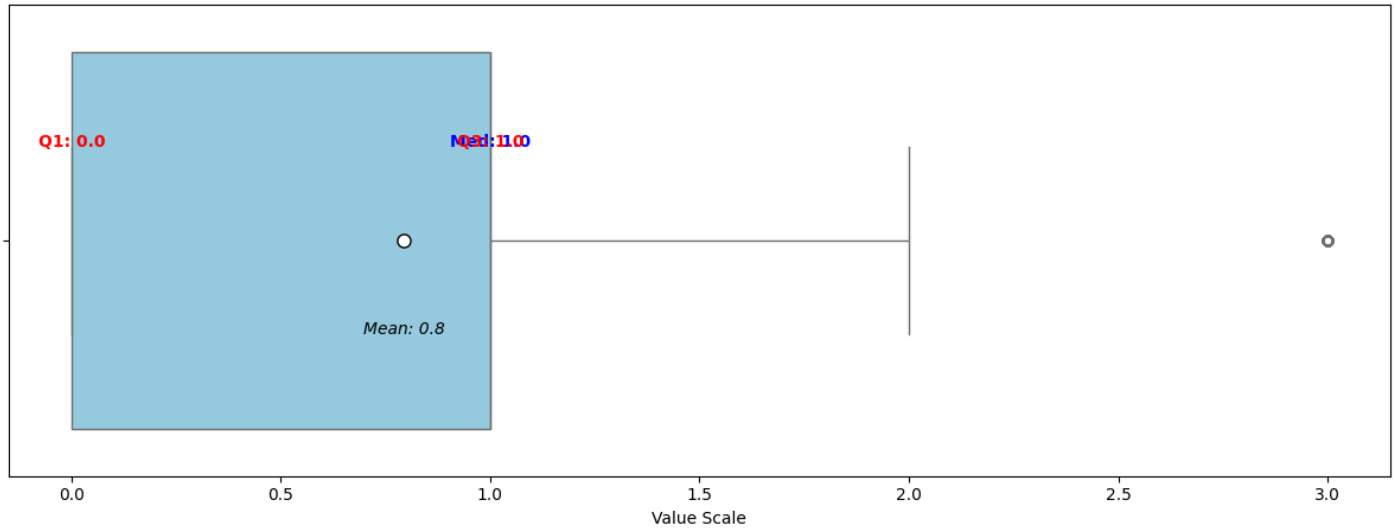
Value Distribution: NumCompaniesWorked



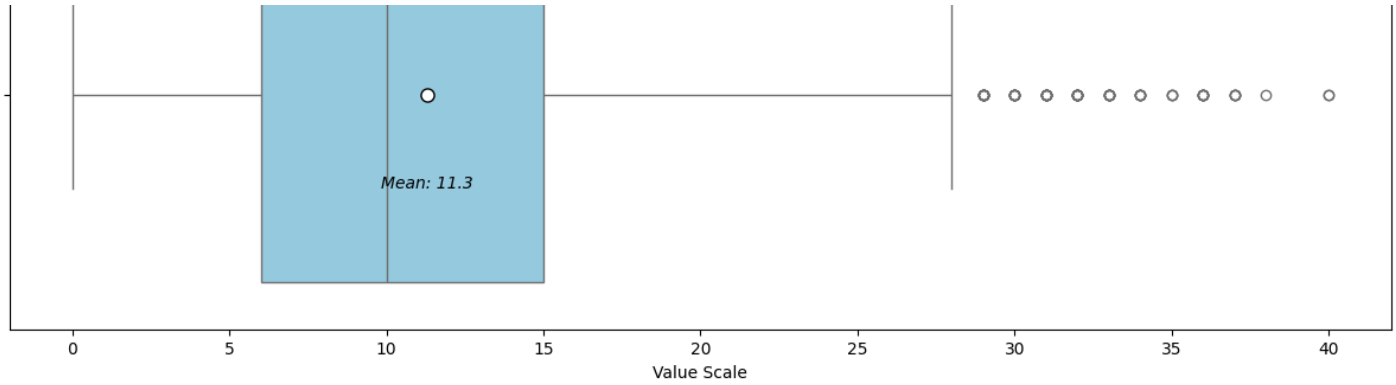
Value Distribution: PerformanceRating



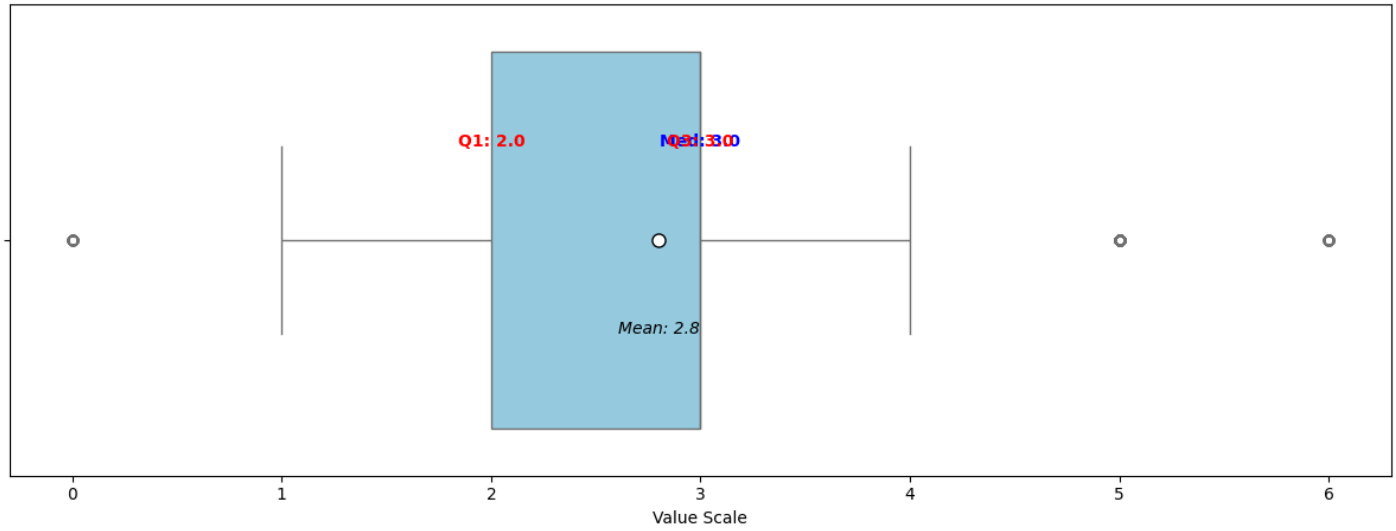
Value Distribution: StockOptionLevel



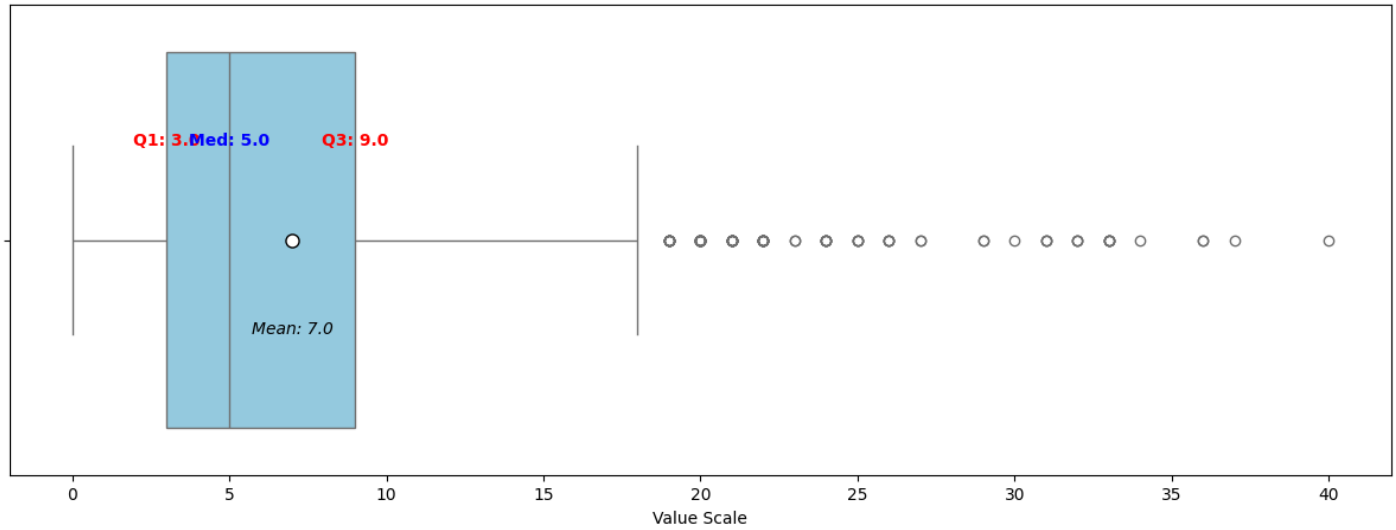
Value Distribution: TotalWorkingYears



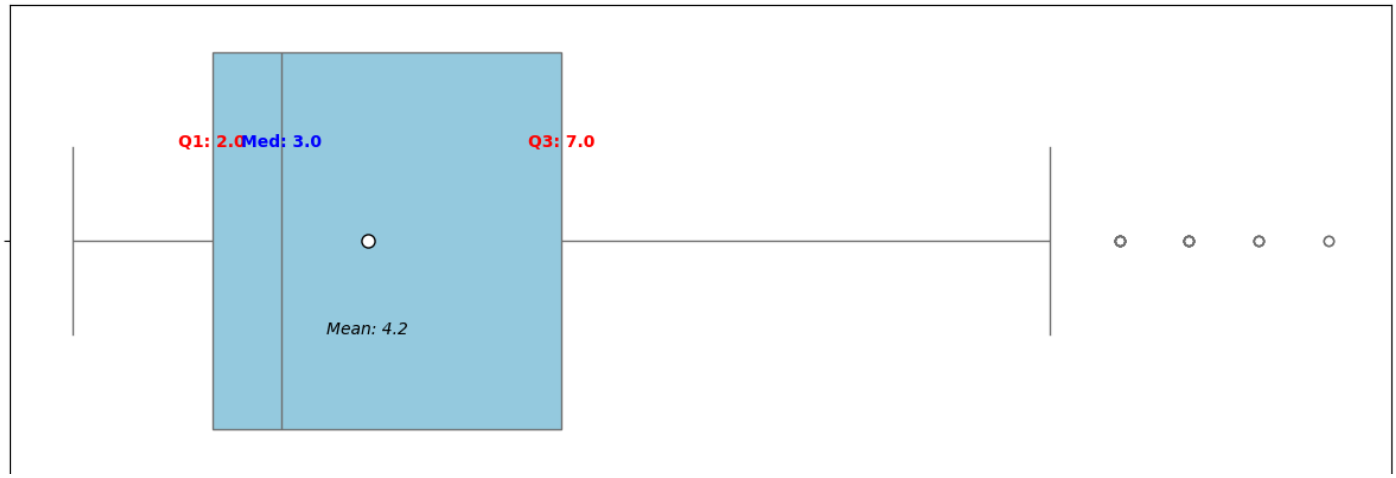
Value Distribution: TrainingTimesLastYear

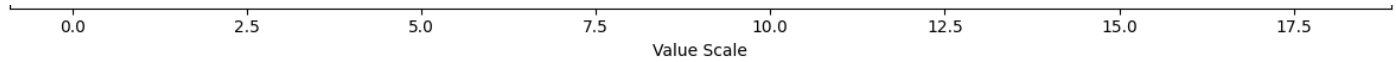


Value Distribution: YearsAtCompany

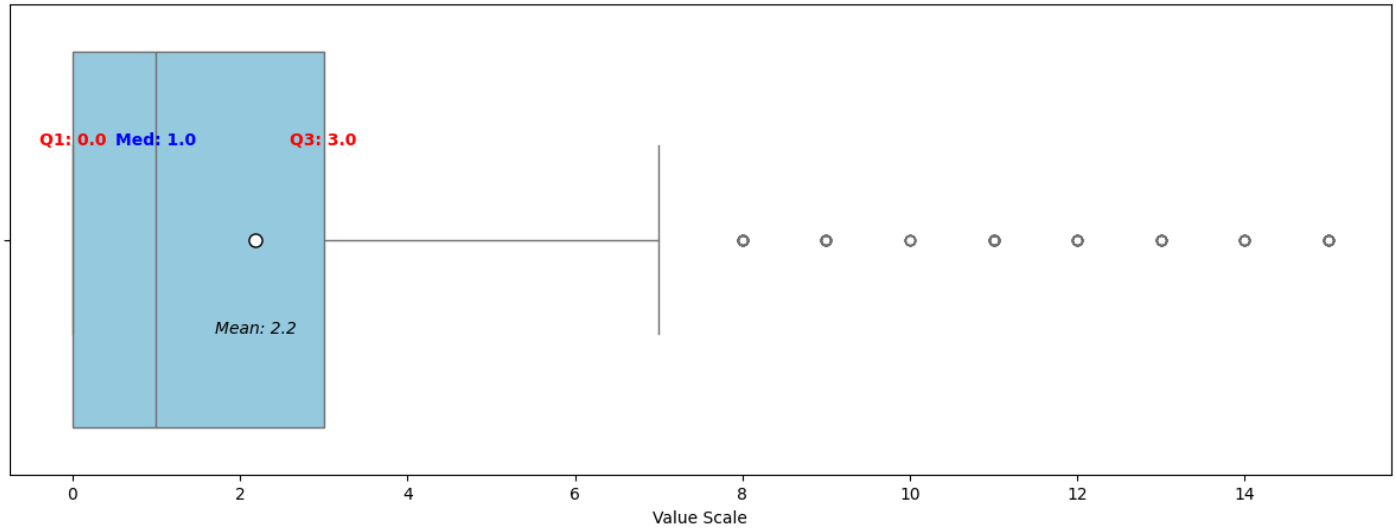


Value Distribution: YearsInCurrentRole

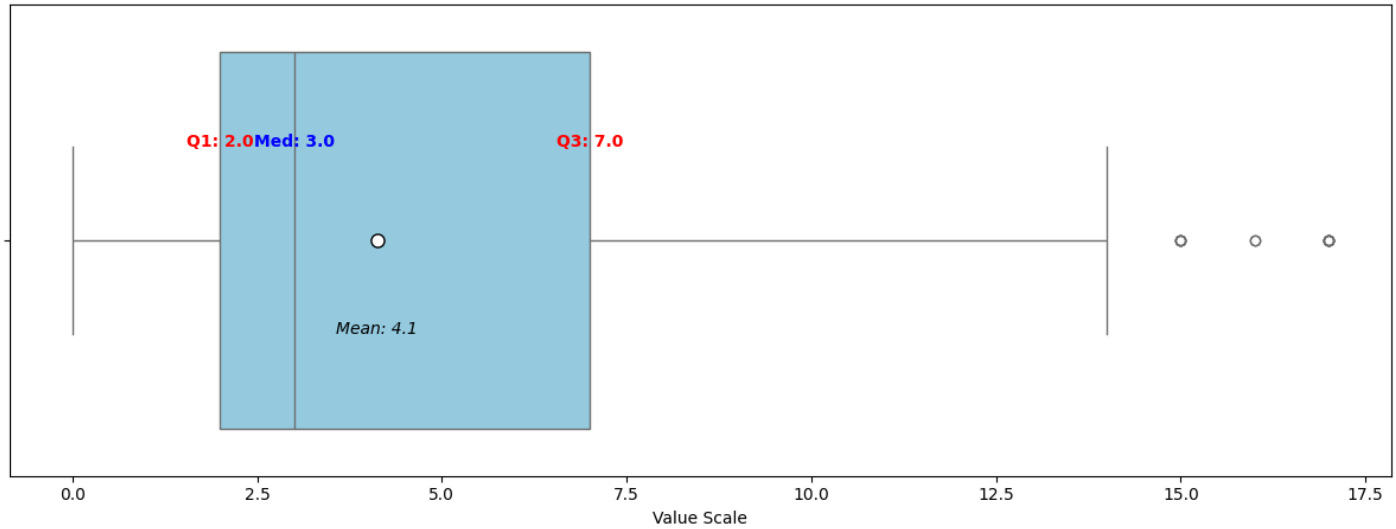




Value Distribution: YearsSinceLastPromotion



Value Distribution: YearsWithCurrManager



	mean	Q1	Q2_Median	Q3	max
MonthlyIncome	6502.931293	2911.0	4919.0	8379.0	19999.0
NumCompaniesWorked	2.693197	1.0	2.0	4.0	9.0
PerformanceRating	3.153741	3.0	3.0	3.0	4.0
StockOptionLevel	0.793878	0.0	1.0	1.0	3.0
TotalWorkingYears	11.279592	6.0	10.0	15.0	40.0
TrainingTimesLastYear	2.799320	2.0	3.0	3.0	6.0
YearsAtCompany	7.008163	3.0	5.0	9.0	40.0
YearsInCurrentRole	4.229252	2.0	3.0	7.0	18.0
YearsSinceLastPromotion	2.187755	0.0	1.0	3.0	15.0
YearsWithCurrManager	4.123129	2.0	3.0	7.0	17.0

In [ ]:

 # Quick health check dataset

```
df.describe().T
```

Out[ ]:

	count	mean	std	min	25%	50%	75%	max
Age	1470.0	36.923810	9.135373	18.0	30.00	36.0	43.00	60.0
DailyRate	1470.0	802.485714	403.509100	102.0	465.00	802.0	1157.00	1499.0
DistanceFromHome	1470.0	9.192517	8.106864	1.0	2.00	7.0	14.00	29.0
Education	1470.0	2.912925	1.024165	1.0	2.00	3.0	4.00	5.0
EmployeeCount	1470.0	1.000000	0.000000	1.0	1.00	1.0	1.00	1.0
EmployeeNumber	1470.0	1024.865306	602.024335	1.0	491.25	1020.5	1555.75	2068.0
EnvironmentSatisfaction	1470.0	2.721769	1.093082	1.0	2.00	3.0	4.00	4.0
HourlyRate	1470.0	65.891156	20.329428	30.0	48.00	66.0	83.75	100.0
JobInvolvement	1470.0	2.729932	0.711561	1.0	2.00	3.0	3.00	4.0
JobLevel	1470.0	2.063946	1.106940	1.0	1.00	2.0	3.00	5.0
JobSatisfaction	1470.0	2.728571	1.102846	1.0	2.00	3.0	4.00	4.0
MonthlyIncome	1470.0	6502.931293	4707.956783	1009.0	2911.00	4919.0	8379.00	19999.0
MonthlyRate	1470.0	14313.103401	7117.786044	2094.0	8047.00	14235.5	20461.50	26999.0
NumCompaniesWorked	1470.0	2.693197	2.498009	0.0	1.00	2.0	4.00	9.0
PercentSalaryHike	1470.0	15.209524	3.659938	11.0	12.00	14.0	18.00	25.0
PerformanceRating	1470.0	3.153741	0.360824	3.0	3.00	3.0	3.00	4.0
RelationshipSatisfaction	1470.0	2.712245	1.081209	1.0	2.00	3.0	4.00	4.0
StandardHours	1470.0	80.000000	0.000000	80.0	80.00	80.0	80.00	80.0
StockOptionLevel	1470.0	0.793878	0.852077	0.0	0.00	1.0	1.00	3.0
TotalWorkingYears	1470.0	11.279592	7.780782	0.0	6.00	10.0	15.00	40.0
TrainingTimesLastYear	1470.0	2.799320	1.289271	0.0	2.00	3.0	3.00	6.0
WorkLifeBalance	1470.0	2.761224	0.706476	1.0	2.00	3.0	3.00	4.0
YearsAtCompany	1470.0	7.008163	6.126525	0.0	3.00	5.0	9.00	40.0
YearsInCurrentRole	1470.0	4.229252	3.623137	0.0	2.00	3.0	7.00	18.0
YearsSinceLastPromotion	1470.0	2.187755	3.222430	0.0	0.00	1.0	3.00	15.0
YearsWithCurrManager	1470.0	4.123129	3.568136	0.0	2.00	3.0	7.00	17.0

Target Variable: Attrition Distribution

Attrition is highly imbalanced:

- No: 1233 employees (83.9%)
- Yes: 237 employees (16.1%)

This imbalance reflects real-world turnover patterns but introduces modeling challenges:

- Models may become biased toward the majority class
- Resampling or class-weighting strategies may be required

Because Attrition is categorical, it is best summarized using counts and percentages.

In [ ]:

```
#Distribution of employees who stayed vs left
%pip install matplotlib
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

# Bar chart
attrition_counts.plot(kind='bar', ax=axes[0], color=['C0', 'C1'])
axes[0].set_title('Employees Stayed vs Left')
axes[0].set_xlabel('Attrition')
axes[0].set_ylabel('Count')
for p in axes[0].patches:
    axes[0].annotate(int(p.get_height()), (p.get_x() + p.get_width() / 2, p.get_height()),
                     ha='center', va='bottom')

# Pie chart
axes[1].pie(attrition_counts, labels=attrition_counts.index, autopct='%1.1f%%',
            colors=['C0', 'C1'], startangle=90, explode=(0, 0.05))
axes[1].set_title('Attrition Proportion')
axes[1].axis('equal')

plt.tight_layout()
plt.show()
```

```
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp312-cp312-win_amd64.whl.metadata (52 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib)
  Downloading fonttools-4.61.1-cp312-cp312-win_amd64.whl.metadata (116 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.4.9-cp312-cp312-win_amd64.whl.metadata (6.4 kB)
Requirement already satisfied: numpy>=1.23 in .\.venv\Lib\site-packages (from matplotlib) (2.4.2)
Requirement already satisfied: packaging>=20.0 in .\.venv\Lib\site-packages (from matplotlib) (26.0)
Collecting pillow>=8 (from matplotlib)
  Downloading pillow-12.1.1-cp312-cp312-win_amd64.whl.metadata (9.0 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil>=2.7 in .\.venv\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in .\.venv\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.10.8-cp312-cp312-win_amd64.whl (8.1 MB)
   ---------------------------------------- 0.0/8.1 MB ? eta -:--:--
   ---------------------- ---------------- 4.7/8.1 MB 23.7 MB/s eta 0:00:01
   ---------------------------------------- 8.1/8.1 MB 24.0 MB/s  0:00:00
Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.61.1-cp312-cp312-win_amd64.whl (2.3 MB)
   ---------------------------------------- 0.0/2.3 MB ? eta -:--:--
   ---------------------------------------- 2.3/2.3 MB 21.9 MB/s  0:00:00
Downloading kiwisolver-1.4.9-cp312-cp312-win_amd64.whl (73 kB)
Downloading pillow-12.1.1-cp312-cp312-win_amd64.whl (7.0 MB)
   ---------------------------------------- 0.0/7.0 MB ? eta -:--:--
   ---------------------------- -------- 5.5/7.0 MB 25.8 MB/s eta 0:00:01
   ---------------------------------------- 7.0/7.0 MB 25.5 MB/s  0:00:00
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools, cycler, contourpy, matplotlib

   ---------------------------------------- 0/7 [pyparsing]
   ----- ---------------------------------- 1/7 [pillow]
   ----- ---------------------------------- 1/7 [pillow]
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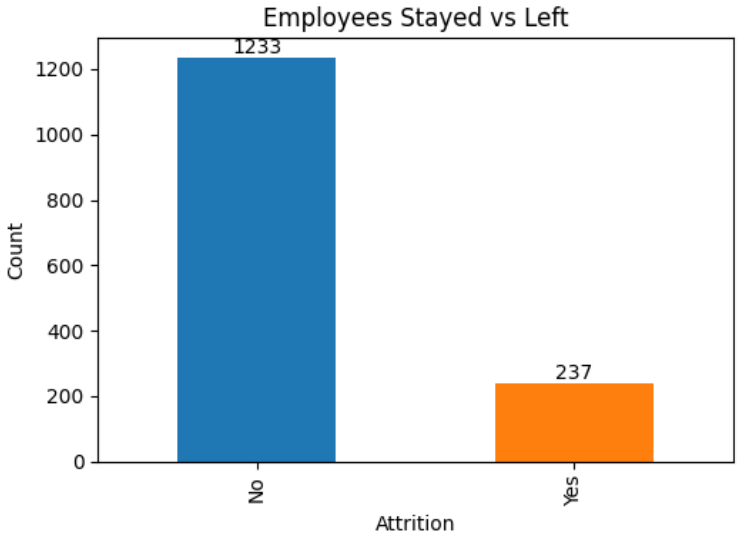
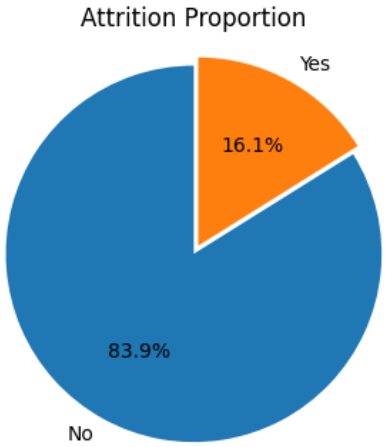
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.61.1 kiwisolver-1.4.9 matplotlib-3.10.8 pillow-12.1.1 pyparsing-3.3.2
Note: you may need to restart the kernel to use updated packages.





Categorical Feature Distributions

Business Travel

- ~71% of employees travel rarely or not at all.

Department & Education Field

- Research & Development dominates (65.4%).
- Most employees have Life Sciences (41.2%) or Medical (31.6%) backgrounds.
- Sales is the second-largest department (30.3%).

Gender

- Workforce is 60% male and 40% female.

Job Roles

Top three roles:

- Sales Executive (22.2%)
- Research Scientist (19.9%)
- Laboratory Technician (17.6%)

Marital Status

- Married: 45.8%
- Single: 32%
- Divorced: 22.2%

Overtime

- Roughly one-third of employees work overtime, a factor often linked to burnout and attrition.

Over18

- Constant "Y" → will be removed during dimensionality reduction.

In [ ]:

```
# Distribution of categorical features with percentage labels (2 charts per row)

charts_per_row = 2
n_cols = len(cat_cols)
n_rows = (n_cols + charts_per_row - 1) // charts_per_row

fig, axes = plt.subplots(n_rows, charts_per_row, figsize=(6 * charts_per_row, 4.5 * n_rows))
axes = axes.flatten()

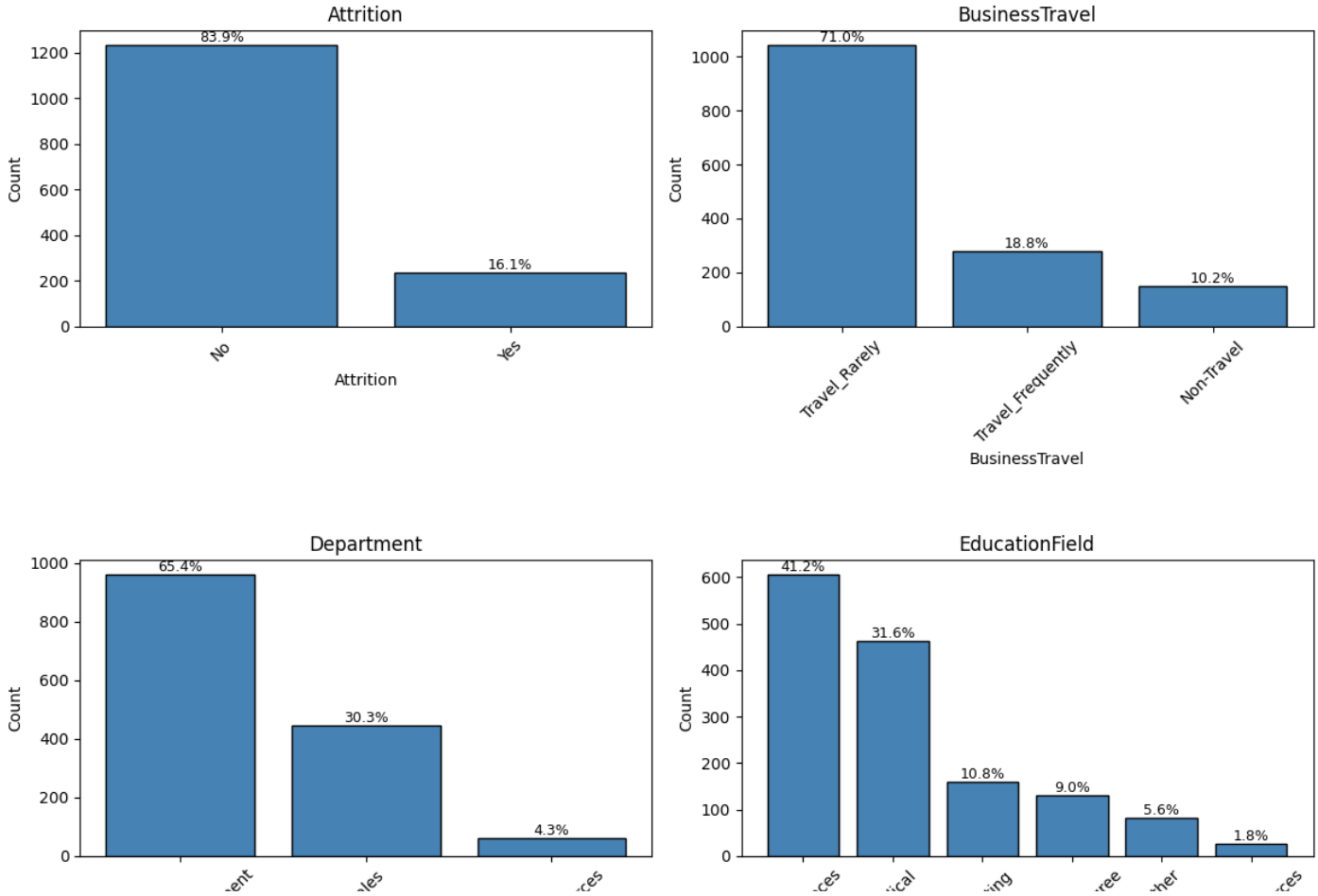
for i, col in enumerate(cat_cols):
    counts = df[col].value_counts(dropna=False)
    percentages = (counts / counts.sum()) * 100

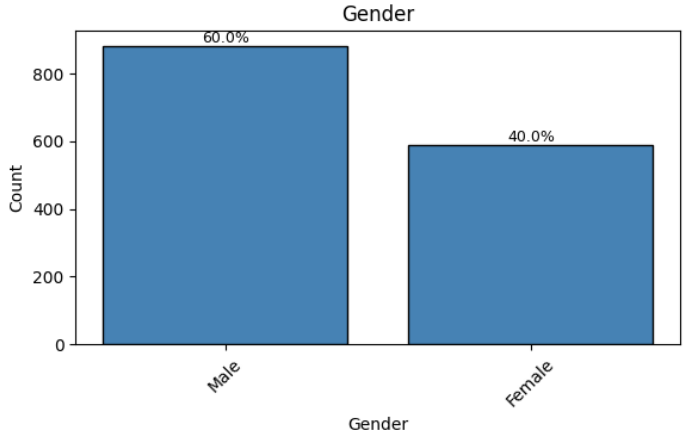
    bars = axes[i].bar(counts.index.astype(str), counts.values, color="steelblue", edgecolor="black")
    axes[i].set_title(col)
    axes[i].set_xlabel(col)
    axes[i].set_ylabel("Count")
    axes[i].tick_params(axis="x", rotation=45)

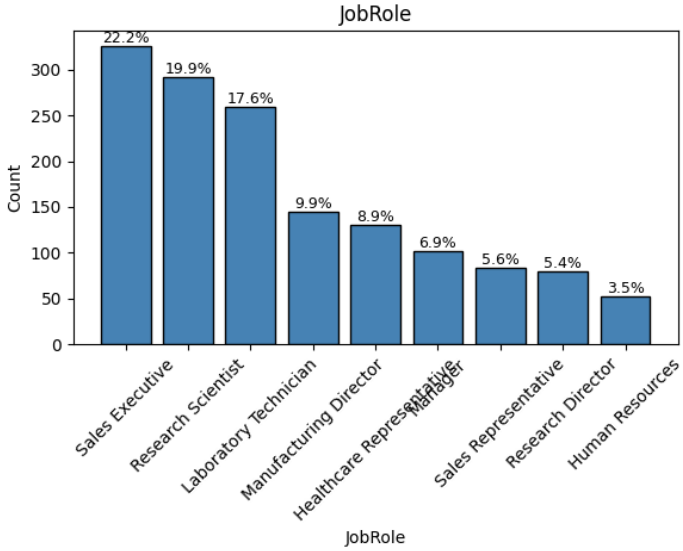
    for bar, pct in zip(bars, percentages.values):
        axes[i].text(
            bar.get_x() + bar.get_width() / 2,
            bar.get_height(),
            f"{pct:.1f}%",
            ha="center",
            va="bottom",
            fontsize=9
        )

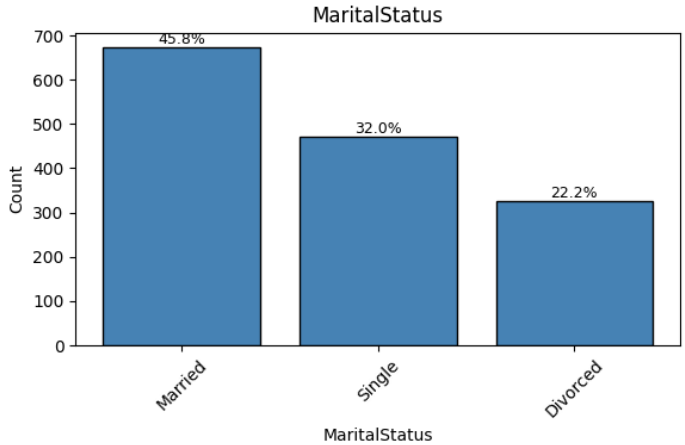
# Hide unused subplot slots
for j in range(i + 1, len(axes)):
    axes[j].axis("off")

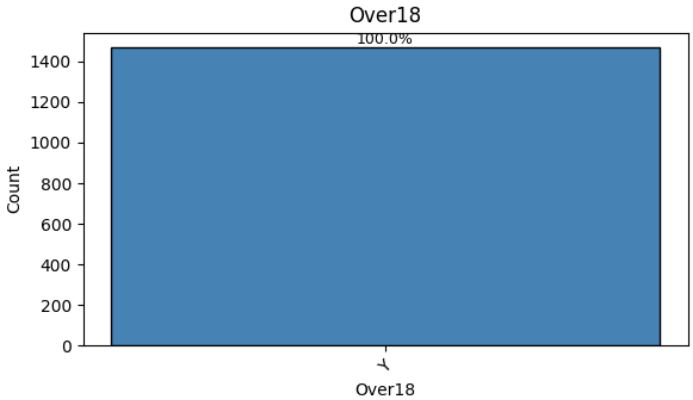
plt.tight_layout()
plt.show()
```

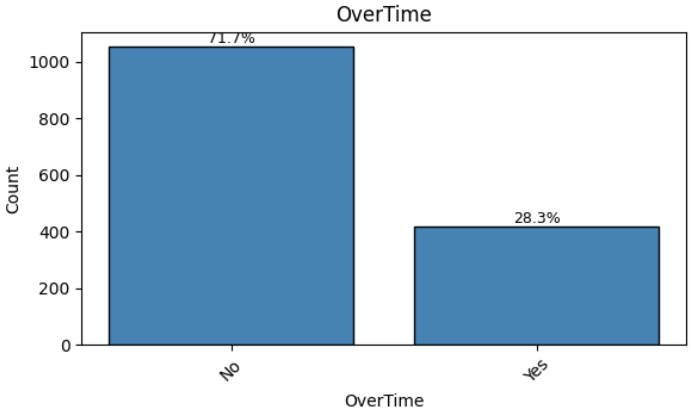












In [ ]:

```
# Print unique values list for each column
for col in df.columns:
    uniques = df[col].unique().tolist()
    print(f"{col} (dtype={df[col].dtype}) - {len(uniques)} unique values")
    print(uniques)
    print("-" * 80)
```

```
Age (dtype=int64) - 43 unique values
[41, 49, 37, 33, 27, 32, 59, 30, 38, 36, 35, 29, 31, 34, 28, 22, 53, 24, 21, 42, 44, 46, 39, 43, 50, 26, 48, 55, 45, 56, 23, 51, 40, 54, 58, 20, 25, 19, 57, 52, 47, 18, 60]
--------------------------------------------------------------------------------
Attrition (dtype=str) - 2 unique values
['Yes', 'No']
```

BusinessTravel (dtype=str) - 3 unique values
['Travel_Rarely', 'Travel_Frequently', 'Non-Travel']

DailyRate (dtype=int64) - 886 unique values
[1102, 279, 1373, 1392, 591, 1005, 1324, 1358, 216, 1299, 809, 153, 670, 1346, 103, 1389, 334, 1123, 1219, 371, 673, 1218, 419, 391, 699, 1282, 1125, 691, 477, 705, 924, 1459, 125
, 895, 813, 1273, 869, 890, 852, 1141, 464, 1240, 1357, 994, 721, 1360, 1065, 408, 1211, 1229, 626, 1434, 1488, 1097, 1443, 515, 853, 1142, 655, 1115, 427, 653, 989, 1435, 1223, 8
36, 1195, 1339, 664, 318, 1225, 1328, 1082, 548, 132, 746, 776, 193, 397, 945, 1214, 111, 573, 1153, 1400, 541, 432, 288, 669, 530, 632, 1334, 638, 1093, 1217, 1353, 120, 682, 489,
807, 827, 871, 665, 1040, 1420, 240, 1280, 534, 1456, 658, 142, 1127, 1031, 1189, 1354, 1467, 922, 394, 1312, 750, 441, 684, 249, 841, 147, 528, 594, 470, 957, 542, 802, 1355, 115
0, 1329, 959, 1033, 1316, 364, 438, 689, 201, 1427, 857, 933, 1181, 1395, 662, 1436, 194, 967, 1496, 1169, 1145, 630, 303, 1256, 440, 1450, 1452, 465, 702, 1157, 602, 1480, 1268,
713, 134, 526, 1380, 140, 629, 1356, 328, 1084, 931, 692, 1069, 313, 894, 556, 1344, 290, 138, 926, 1261, 472, 1002, 878, 905, 1180, 121, 1136, 635, 1151, 644, 1045, 829, 1242, 14
69, 896, 992, 1052, 1147, 1396, 663, 119, 979, 319, 1413, 944, 1323, 532, 818, 854, 1034, 771, 1401, 1431, 976, 1411, 1300, 252, 1327, 832, 1017, 1199, 504, 505, 916, 1247, 685, 2
69, 1416, 833, 307, 1311, 128, 488, 529, 1210, 1463, 675, 1385, 1403, 452, 666, 1158, 228, 996, 728, 1315, 322, 1479, 797, 1070, 442, 496, 1372, 920, 688, 1449, 1117, 636, 506, 44
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Department (dtype=str) - 3 unique values
['Sales', 'Research & Development', 'Human Resources']

DistanceFromHome (dtype=int64) - 29 unique values
[1, 8, 2, 3, 24, 23, 27, 16, 15, 26, 19, 21, 5, 11, 9, 7, 6, 10, 4, 25, 12, 18, 29, 22, 14, 20, 28, 17, 13]

Education (dtype=int64) - 5 unique values
[2, 1, 4, 3, 5]

EducationField (dtype=str) - 6 unique values
['Life Sciences', 'Other', 'Medical', 'Marketing', 'Technical Degree', 'Human Resources']

EmployeeCount (dtype=int64) - 1 unique values
[1]

EmployeeNumber (dtype=int64) - 1470 unique values
[1, 2, 4, 5, 7, 8, 10, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22, 23, 24, 26, 27, 28, 30, 31, 32, 33, 35, 36, 38, 39, 40, 41, 42, 45, 46, 47, 49, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 62, 63, 64
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1549, 1550, 1551, 1552, 1553, 1554, 1555, 1556, 1557, 1558, 1560, 1562, 1563, 1564, 1568, 1569, 1572, 1573, 1574, 1576, 1577, 1578, 1580, 1581, 1582, 1583, 1585, 1586, 1587, 1

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1701, 1702, 1703, 1704, 1706, 1707, 1708, 1709, 1710, 1712, 1714, 1716, 1718, 1719, 1720, 1721, 1722, 1724, 1725, 1727, 1728, 1729, 1731, 1732, 1733, 1734, 1735, 1736, 1737, 1
739, 1740, 1744, 1745, 1746, 1747, 1749, 1751, 1752, 1753, 1754, 1755, 1756, 1757, 1758, 1760, 1761, 1762, 1763, 1764, 1766, 1767, 1768, 1770, 1771, 1772, 1774, 1775, 1778, 17
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1865, 1866, 1867, 1868, 1869, 1870, 1871, 1873, 1875, 1876, 1878, 1880, 1881, 1882, 1883, 1885, 1886, 1888, 1890, 1892, 1893, 1898, 1900, 1903, 1905, 1907, 1908, 1909, 1911, 1
912, 1915, 1916, 1918, 1922, 1924, 1927, 1928, 1929, 1931, 1932, 1933, 1934, 1935, 1936, 1937, 1938, 1939, 1940, 1941, 1943, 1944, 1945, 1947, 1948, 1949, 1950, 1951, 1952, 19
54, 1955, 1956, 1960, 1961, 1962, 1965, 1966, 1967, 1968, 1969, 1970, 1971, 1972, 1973, 1974, 1975, 1976, 1979, 1980, 1981, 1982, 1985, 1986, 1987, 1989, 1992, 1993, 1994, 199
5, 1996, 1997, 1998, 1999, 2000, 2003, 2007, 2008, 2009, 2010, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2031, 2032, 2034,
2035, 2036, 2037, 2038, 2040, 2041, 2044, 2045, 2046, 2048, 2049, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2060, 2061, 2062, 2064, 2065, 2068]

EnvironmentSatisfaction (dtype=int64) - 4 unique values
[2, 3, 4, 1]

Gender (dtype=str) - 2 unique values
['Female', 'Male']

HourlyRate (dtype=int64) - 71 unique values
[94, 61, 92, 56, 40, 79, 81, 67, 44, 84, 49, 31, 93, 50, 51, 80, 96, 78, 45, 82, 53, 83, 58, 72, 48, 42, 41, 86, 97, 75, 33, 37, 73, 98, 36, 47, 71, 30, 43, 99, 59, 95, 57, 76, 87, 66, 55, 32, 52
, 70, 62, 64, 63, 60, 100, 46, 39, 77, 35, 91, 54, 34, 90, 65, 88, 85, 89, 68, 69, 74, 38]

JobInvolvement (dtype=int64) - 4 unique values
[3, 2, 4, 1]

JobLevel (dtype=int64) - 5 unique values
[2, 1, 3, 4, 5]

JobRole (dtype=str) - 9 unique values
['Sales Executive', 'Research Scientist', 'Laboratory Technician', 'Manufacturing Director', 'Healthcare Representative', 'Manager', 'Sales Representative', 'Research Dire
ctor', 'Human Resources']

JobSatisfaction (dtype=int64) - 4 unique values
[4, 2, 3, 1]

MaritalStatus (dtype=str) - 3 unique values
['Single', 'Married', 'Divorced']

MonthlyIncome (dtype=int64) - 1349 unique values
[5993, 5130, 2090, 2909, 3468, 3068, 2670, 2693, 9526, 5237, 2426, 4193, 2911, 2661, 2028, 9980, 3298, 2935, 15427, 3944, 4011, 3407, 11994, 1232, 2960, 19094, 3919, 6825, 102
48, 18947, 2496, 6465, 2206, 2086, 2293, 2645, 2683, 2014, 3419, 5376, 1951, 2341, 8726, 19545, 4568, 3022, 5772, 2269, 5381, 3441, 5454, 9884, 4157, 13458, 9069, 4014, 5915,
6162, 2406, 18740, 7637, 10096, 14756, 6499, 9724, 2194, 3388, 5473, 2703, 2501, 6220, 3038, 4424, 4312, 13245, 13664, 5021, 5126, 2859, 10239, 5329, 4325, 7260, 2322, 2075,
4152, 9619, 13503, 5441, 5209, 10673, 5010, 13549, 4999, 4221, 13872, 2042, 2073, 2956, 2926, 4809, 5163, 18844, 18172, 5744, 2889, 2871, 7484, 6074, 17328, 2774, 4505, 7428,
11631, 9738, 2835, 16959, 2613, 6146, 4963, 19537, 6172, 2368, 10312, 1675, 2523, 6567, 4739, 9208, 4559, 8189, 2942, 4941, 10650, 5902, 8639, 6347, 4200, 3452, 4317, 2632, 4
668, 3204, 2720, 17181, 2238, 1483, 5605, 7295, 2306, 2348, 8998, 4319, 6132, 3346, 10855, 2231, 2323, 2024, 2713, 9439, 2566, 19926, 2451, 9419, 8686, 3058, 2325, 2088, 3072,
5006, 4257, 2500, 1102, 10453, 2288, 3929, 2311, 3140, 3690, 4450, 2756, 19033, 18722, 9547, 13734, 19999, 2279, 5916, 2089, 16792, 3564, 4425, 5265, 6553, 6261, 4298, 6804,
3815, 2741, 6673, 7639, 2328, 2153, 4876, 9396, 10400, 8474, 9981, 12490, 2657, 13591, 6696, 2058, 8865, 5940, 5914, 2622, 12185, 10609, 4345, 2177, 2793, 7918, 8789, 2389, 3
212, 19232, 2267, 19517, 2436, 16064, 2707, 19068, 3931, 3730, 2232, 4465, 3319, 19202, 13675, 5957, 3920, 6434, 10048, 10938, 2340, 6545, 6931, 4898, 2593, 19436, 2723, 347
9, 2794, 5249, 2176, 16872, 3485, 6644, 5582, 4000, 13496, 3210, 19045, 11849, 2070, 6502, 3230, 13603, 11996, 6397, 19144, 17584, 4907, 4554, 5415, 4741, 2115, 3161, 5745, 2
373, 3310, 18665, 4485, 2789, 5828, 2326, 13525, 1420, 8020, 3688, 5482, 16015, 1200, 5661, 6929, 9613, 5674, 5484, 12061, 5660, 4821, 6410, 5210, 2695, 11878, 17068, 2455, 1
3964, 2478, 5228, 4478, 7547, 5055, 3464, 5775, 8943, 19272, 5238, 4682, 18300, 5257, 6349, 4869, 9985, 3697, 7457, 2119, 3983, 6118, 6214, 11510, 7143, 8268, 8095, 2904, 603
2, 2976, 15992, 4649, 2696, 2370, 12504, 5974, 4736, 5296, 6781, 2174, 6653, 9699, 6755, 2213, 2610, 2851, 5258, 9355, 10496, 6380, 2716, 2201, 6540, 3816, 5253, 10965, 4936,
2543, 5304, 16659, 4260, 2476, 3102, 2244, 7596, 2285, 3034, 5715, 2576, 4197, 14336, 3448, 19406, 6538, 4306, 2258, 4522, 4487, 4449, 2218, 19197, 13212, 6577, 8392, 4558, 4
031, 7969, 2654, 16555, 4556, 6091, 19566, 4810, 4523, 3202, 2351, 1702, 18041, 2886, 2097, 11935, 2546, 2564, 8412, 14118, 17046, 10266, 5070, 17861, 4230, 3780, 2768, 9071,
10648, 13610, 3408, 2983, 7632, 9824, 9950, 2093, 3894, 4051, 16835, 6230, 4717, 13237, 3755, 6582, 7406, 4805, 4262, 16184, 11557, 1878, 10932, 6811, 4859, 5337, 7491, 1052
7, 16595, 8834, 5577, 4707, 2400, 6447, 19502, 2725, 6272, 2127, 18200, 2096, 2033, 3622, 4233, 3681, 5460, 2187, 9602, 2836, 4089, 16627, 2619, 5679, 15402, 5985, 2579, 3041,
3447, 19513, 2773, 7104, 6322, 2083, 8381, 2691, 4286, 2659, 9434, 5561, 6646, 7725, 10725, 8847, 2045, 1009, 3348, 1281, 2819, 4851, 4028, 8120, 4647, 4680, 3221, 8621, 4577,
4553, 5396, 6796, 7625, 7412, 11159, 4960, 10475, 14814, 19141, 5405, 8793, 19189, 3875, 2216, 11713, 7861, 3708, 13770, 2642, 2759, 6142, 6389, 11103, 2342, 2297, 2450, 509
3, 5309, 3057, 5121, 16856, 2686, 6180, 6632, 3505, 6274, 19859, 7587, 4258, 4364, 4335, 5326, 3280, 5485, 4342, 2782, 5980, 4381, 2572, 3833, 4244, 6500, 18430, 1601, 2694, 3
149, 17639, 2319, 11691, 5324, 16752, 2700, 19246, 2506, 6062, 4382, 2143, 5094, 6877, 2274, 4434, 6288, 2553, 7654, 5160, 17159, 12808, 10221, 4779, 3737, 2366, 1706, 16307,
5933, 3424, 4037, 2559, 6201, 4403, 3761, 10934, 10761, 5175, 13826, 6334, 4775, 2818, 2515, 4194, 10685, 2022, 2314, 4256, 3580, 3162, 6524, 2899, 5231, 2356, 2800, 11836, 1
0903, 2973, 14275, 5562, 4537, 7642, 17924, 5204, 2277, 2795, 2532, 4908, 2380, 4765, 2044, 6586, 3294, 4171, 2778, 2377, 2404, 2318, 2008, 6244, 2799, 10552, 2329, 7403, 225
9, 6932, 4678, 13582, 2332, 2413, 9705, 4294, 4721, 2519, 2121, 5855, 3617, 6725, 10325, 6949, 4447, 2157, 4601, 17099, 2479, 14852, 7264, 5666, 7823, 7880, 13194, 5067, 5079,
2321, 17444, 2270, 17399, 5488, 19419, 2811, 3633, 4163, 2132, 13973, 2684, 10845, 4377, 3743, 4148, 1051, 10739, 10388, 11416, 2600, 2422, 5472, 4240, 10999, 5003, 12742, 4
227, 3917, 18303, 13726, 4777, 6385, 19973, 6861, 4969, 19845, 13320, 2743, 10880, 17650, 4025, 9725, 11904, 7525, 4834, 2220, 1052, 2821, 19237, 4107, 8396, 2007, 19627, 10
686, 8858, 16756, 10798, 1416, 4615, 2461, 8722, 3955, 9957, 3376, 8823, 10322, 4621, 10976, 3660, 10482, 7119, 9582, 4508, 2207, 7756, 6694, 3691, 2313, 17665, 2596, 4728, 4
302, 2979, 16885, 5593, 10445, 8740, 2514, 7655, 17465, 7351, 10820, 12169, 19626, 6782, 7779, 2791, 3201, 4968, 13120, 4033, 3291, 4272, 5056, 2844, 1904, 8224, 4766, 5731,
2539, 5714, 4323, 7336, 13499, 13758, 5155, 3597, 4420, 6578, 4422, 10274, 5343, 2376, 5346, 2827, 19943, 3131, 2552, 4477, 6474, 3033, 2936, 18606, 2168, 2853, 17048, 2290,
3600, 2107, 4115, 4327, 17856, 3196, 19081, 8966, 2210, 4539, 3491, 4541, 2678, 7379, 5220, 4998, 10252, 2781, 6852, 4950, 3579, 13191, 10377, 2235, 10502, 2011, 1859, 3760,
17779, 6833, 6812, 5171, 19740, 18711, 3692, 2517, 6623, 18265, 16124, 2585, 18213, 8380, 2994, 1223, 1118, 2875, 18824, 13577, 2625, 18789, 4538, 19847, 10512, 4444, 2154,
19190, 4490, 3506, 2372, 10231, 5410, 7978, 3867, 2838, 4695, 3339, 2080, 6209, 18061, 17123, 4883, 3904, 4627, 7094, 3423, 6674, 16880, 9094, 8446, 11916, 4534, 9852, 6151,
2302, 2362, 19187, 19717, 3544, 8500, 4661, 4103, 4249, 14026, 6893, 6125, 3669, 10008, 2387, 4639, 7898, 2534, 13142, 1611, 5363, 5071, 13695, 13402, 2029, 6377, 5429, 2785,
4614, 6687, 4724, 6179, 6120, 10596, 5467, 2996, 9998, 4078, 10920, 6232, 13247, 4081, 5769, 2394, 16799, 2950, 3629, 9362, 3229, 3578, 7988, 4284, 7553, 19701, 14732, 9278,
1359, 16422, 1261, 2099, 5810, 5647, 3420, 4400, 3500, 2066, 17169, 4162, 9204, 3975, 10793, 3646, 7446, 10851, 2109, 3722, 9380, 5486, 2742, 13757, 8463, 16598, 6651, 2345,
4373, 4759, 5301, 3673, 4768, 1274, 4900, 10466, 17007, 5765, 4599, 3172, 10209, 8620, 2064, 4035, 3838, 4591, 2561, 1563, 4789, 3180, 6549, 6388, 11244, 16032, 16328, 8376,
16606, 8606, 2272, 2018, 7083, 4084, 14411, 2308, 4841, 4285, 9715, 4320, 10124, 5207, 16437, 2296, 4069, 7441, 2430, 5878, 2644, 6439, 6392, 9714, 6077, 9250, 2074, 10169, 4
855, 4087, 2367, 2972, 19586, 2061, 9924, 4198, 6815, 4723, 8237, 8853, 19331, 19613, 5063, 2690, 17567, 2408, 2814, 11245, 3312, 19049, 2141, 4385, 5332, 4663, 3211, 5377, 4
066, 5208, 4877, 3117, 1569, 19658, 3069, 10435, 5768, 5042, 5770, 10306, 3936, 7945, 5743, 15202, 5440, 3517, 2580, 2166, 5869, 8008, 5206, 5295, 16413, 13269, 2783, 5433, 2
013, 13966, 4374, 6842, 17426, 17603, 4581, 4735, 4187, 5505, 5470, 5476, 2587, 2440, 15972, 15379, 7082, 2728, 5368, 5347, 3195, 3989, 3306, 7005, 2655, 1393, 2570, 3537, 39
86, 10883, 9525, 2929, 2275, 7879, 4930, 7847, 4401, 9241, 2974, 4502, 10748, 1555, 12936, 2305, 16704, 3433, 3477, 6430, 6516, 3907, 6883, 2862, 4978, 10368, 6134, 6735, 329
5, 6472, 9610, 19833, 9756, 2145, 2180, 8346, 3445, 2760, 6294, 7140, 2932, 5147, 4507, 8564, 2468, 8161, 5294, 2718, 5811, 2437, 2766, 19038, 3055, 2289, 4001, 12965, 3539, 6

029, 2679, 3702, 2398, 5468, 13116, 4189, 19328, 8321, 4071, 5813, 3143, 13464, 7991, 3377, 5538, 5762, 2592, 4213, 4127, 2438, 6870, 10447, 9667, 2148, 8926, 6513, 6799, 16291, 2705, 10333, 4448, 6854, 9637, 3591, 4684, 15787, 1514, 2335, 5154, 6962, 5675, 2379, 3812, 4648, 2105, 8578, 2706, 6384, 3968, 9907, 13225, 3540, 2804, 19392, 19665, 2439, 7314, 4774, 3902, 2662, 2856, 1081, 2472, 5673, 9713, 2062, 4788, 5906, 3886, 16823, 2933, 17174, 5033, 2307, 5507, 4393, 13348, 6583, 8103, 3978, 2544, 5399, 5487, 6834, 1091, 5736, 2226, 5747, 9854, 5380, 5151, 2133, 17875, 2432, 4771, 19161, 5087, 2863, 2144, 3065, 2810, 9888, 8628, 2867, 5373, 6667, 2858, 4105, 9679, 5617, 10448, 2897, 5968, 7510, 2991, 19636, 1129, 13341, 4332, 11031, 4440, 4617, 2647, 6323, 5677, 3748, 3977, 8633, 3067, 5321, 11957, 2660, 3375, 5098, 4878, 2837, 4108, 13206, 10422, 13744, 3482, 19431, 1790, 7644, 5131, 6306, 4787, 18880, 2339, 13570, 6712, 5406, 8938, 8837, 6728, 6652, 4850, 2809, 5689, 2001, 2977, 3785, 10854, 12031, 9936, 2966, 2571, 9991, 5390, 4404]

MonthlyRate (dtype=int64) - 1427 unique values
[19479, 24907, 2396, 23159, 16632, 11864, 9964, 13335, 8787, 16577, 16479, 12682, 15170, 8758, 12947, 10195, 15053, 7324, 22021, 4306, 8232, 6986, 21293, 19281, 17102, 10735, 4681, 21173, 2094, 22822, 6670, 19121, 16117, 3335, 3020, 21923, 3810, 9687, 13072, 3193, 10910, 19715, 10558, 2975, 10781, 16280, 10034, 10227, 20445, 4892, 19294, 11179, 4009, 8302, 21436, 15146, 11031, 16002, 9528, 2689, 10877, 5456, 16701, 2354, 8202, 19730, 22656, 18787, 5868, 21777, 24668, 4956, 18775, 7346, 22002, 20682, 23016, 15067, 25258, 10425, 15998, 26278, 18092, 15717, 17736, 21698, 9518, 18725, 15830, 13596, 14115, 8423, 19760, 3142, 24301, 24001, 17519, 8863, 24409, 25043, 23648, 15178, 19783, 12482, 15850, 21922, 9755, 26959, 26897, 23785, 25796, 22887, 13871, 13257, 15000, 14506, 5615, 22952, 2561, 19494, 22310, 15480, 4510, 6462, 20739, 23300, 3465, 26820, 19299, 5549, 16090, 6645, 24788, 21196, 8916, 2819, 25150, 14590, 24835, 13982, 10224, 9752, 2302, 23910, 22812, 10415, 11162, 12888, 6961, 16102, 8504, 11439, 16047, 10901, 15589, 26283, 13983, 11873, 8552, 11314, 11992, 5970, 6672, 23402, 25326, 17053, 4609, 8053, 12930, 12430, 13364, 20989, 15062, 11012, 6319, 13939, 10515, 9241, 2137, 6984, 5711, 21728, 3425, 26250, 4673, 6499, 13339, 14074, 7192, 5678, 11781, 15497, 5228, 20462, 22977, 15986, 16439, 7259, 4185, 9679, 23793, 5972, 16523, 11354, 24525, 12392, 7703, 14242, 12368, 25812, 20925, 12916, 15736, 8556, 14674, 22967, 19757, 16840, 17011, 9945, 13248, 10056, 9647, 4381, 8456, 2539, 6599, 9096, 14961, 3300, 4933, 25657, 24118, 22149, 7744, 21509, 21030, 20990, 9571, 15417, 12069, 19877, 24447, 15970, 13523, 14776, 23687, 18697, 5118, 22573, 6420, 22673, 10732, 7505, 8007, 5949, 23231, 11652, 26062, 19682, 19737, 14977, 14935, 3687, 14408, 18384, 7501, 20251, 18938, 10268, 2613, 22825, 10531, 11677, 19100, 19191, 26767, 15815, 21016, 24532, 20260, 15972, 22722, 15881, 19920, 18899, 14180, 4488, 25594, 26285, 3909, 8450, 11411, 14864, 25233, 5100, 7122, 16321, 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NumCompaniesWorked (dtype=int64) - 10 unique values
[8, 1, 6, 9, 0, 4, 5, 2, 7, 3]

Over18 (dtype=str) - 1 unique values
['Y']

OverTime (dtype=str) - 2 unique values
['Yes', 'No']

PercentSalaryHike (dtype=int64) - 15 unique values
[11, 23, 15, 12, 13, 20, 22, 21, 17, 14, 16, 18, 19, 24, 25]

PerformanceRating (dtype=int64) - 2 unique values
[3, 4]

RelationshipSatisfaction (dtype=int64) - 4 unique values

```
RelationshipSatisfaction (dtype=int64) - 4 unique values
[1, 4, 2, 3]
--------------------------------------------------------------------------------
StandardHours (dtype=int64) - 1 unique values
[80]
--------------------------------------------------------------------------------
StockOptionLevel (dtype=int64) - 4 unique values
[0, 1, 3, 2]
--------------------------------------------------------------------------------
TotalWorkingYears (dtype=int64) - 40 unique values
[8, 10, 7, 6, 12, 1, 17, 5, 3, 31, 13, 0, 26, 24, 22, 9, 19, 2, 23, 14, 15, 4, 29, 28, 21, 25, 20, 11, 16, 37, 38, 30, 40, 18, 36, 34, 32, 33, 35, 27]
--------------------------------------------------------------------------------
TrainingTimesLastYear (dtype=int64) - 7 unique values
[0, 3, 2, 5, 1, 4, 6]
--------------------------------------------------------------------------------
WorkLifeBalance (dtype=int64) - 4 unique values
[1, 3, 2, 4]
--------------------------------------------------------------------------------
YearsAtCompany (dtype=int64) - 37 unique values
[6, 10, 0, 8, 2, 7, 1, 9, 5, 4, 25, 3, 12, 14, 22, 15, 27, 21, 17, 11, 13, 37, 16, 20, 40, 24, 33, 19, 36, 18, 29, 31, 32, 34, 26, 30, 23]
--------------------------------------------------------------------------------
YearsInCurrentRole (dtype=int64) - 19 unique values
[4, 7, 0, 2, 5, 9, 8, 3, 6, 13, 1, 15, 14, 16, 11, 10, 12, 18, 17]
--------------------------------------------------------------------------------
YearsSinceLastPromotion (dtype=int64) - 16 unique values
[0, 1, 3, 2, 7, 4, 8, 6, 5, 15, 9, 13, 12, 10, 11, 14]
--------------------------------------------------------------------------------
YearsWithCurrManager (dtype=int64) - 18 unique values
[5, 7, 0, 2, 6, 8, 3, 11, 17, 1, 4, 12, 9, 10, 15, 13, 16, 14]
--------------------------------------------------------------------------------
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Summary of Key Univariate EDA Findings

- Dataset is clean and ready for modeling
- Several numerical features are right-skewed
- Multiple features show zero variance and should be removed
- Outliers represent meaningful employee subgroups
- Attrition is highly imbalanced (84% vs 16%)
- Workforce is dominated by R&D roles and Life Sciences backgrounds
- Overtime, job role, tenure, and income patterns show potential relationships with attrition
- These insights provide a strong foundation for correlation analysis, feature engineering, and predictive modeling.